Rapid Microstructure Prediction in Laser Powder Bed Fusion Using Bayesian Updating of Eagar–Tsai Model

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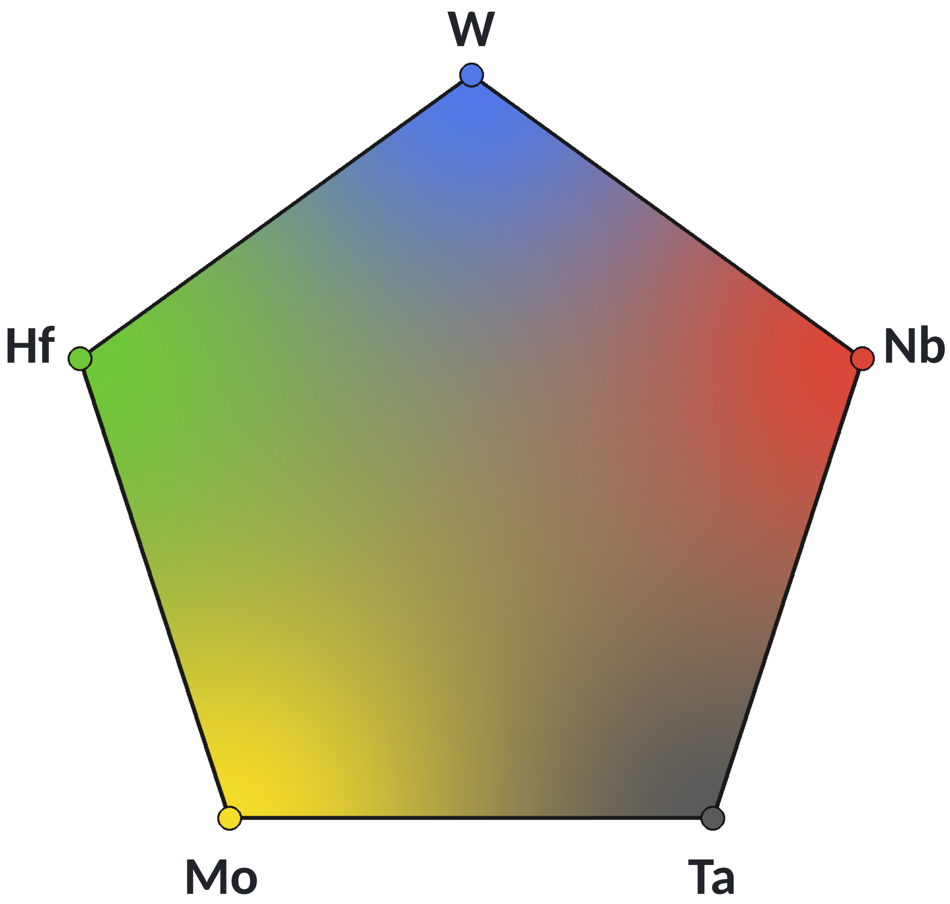
The Design Space Is Large



Five-element projection of a multicomponent composition space.

Each alloy has many possible process conditions

Composition–Process Space

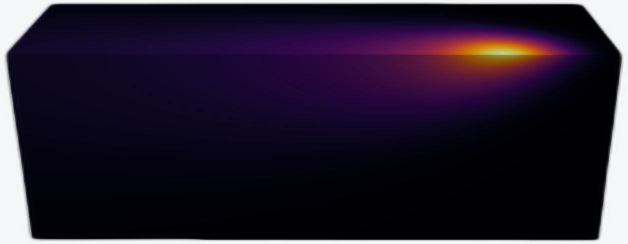
Composition $\times P \times v \times A$

- Cannot practically evaluate every combination with high-fidelity calculations.
- Rapid thermal prediction is needed for high-throughput screening.

Core Thesis

ET is fast enough for high-throughput screening, but not yet reliable enough for microstructure decisions.

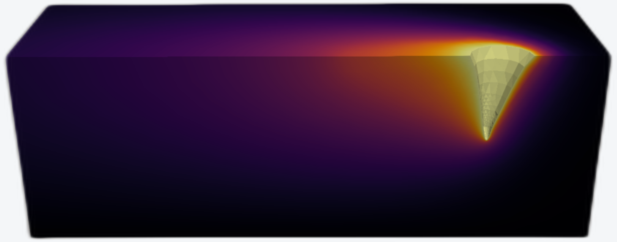
Eagar–Tsai



- Fast, but has simplified physics
- Run across the design space

+

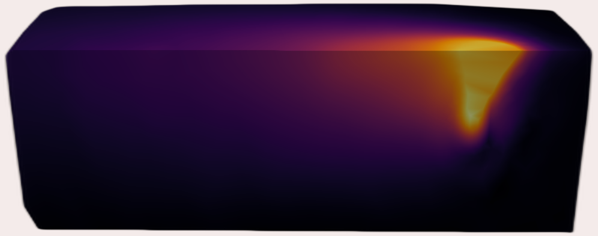
Thermo-Calc AM Module



- Higher fidelity, but more computationally expensive
- Run at selected conditions

$\longrightarrow$

Bayesian-Updated ET



- Learn the TCAM–ET discrepancy
- Correct the ET temperature field

Microstructure Prediction

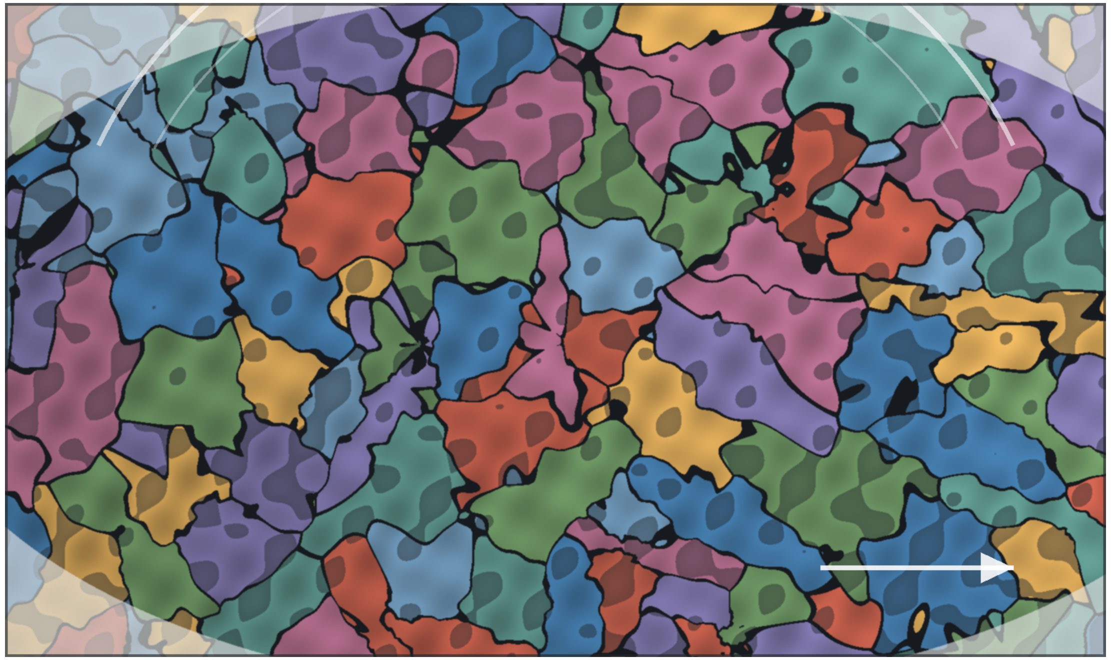
Corrected 3D temperature field $\longrightarrow$ G and R $\longrightarrow$ KGT $\longrightarrow$ As-solidified microstructure

Why Predict Microstructure?

Process parameters affect performance through the thermal history and resulting microstructure.

Composition + Process $\longrightarrow$ **Thermal History** $\longrightarrow$ **Microstructure**

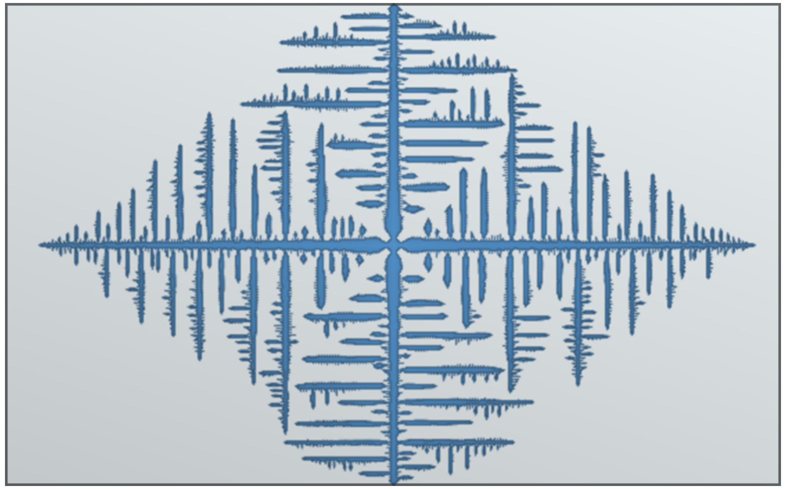
- Microstructure controls strength, ductility, anisotropy, and defect sensitivity.
- A sound melt pool can still solidify into the wrong morphology.
- Predicting structure reduces trial-and-error printing.
- Fast predictions enable composition and process screening.



Illustrative phase-field style microstructure proxy: thermal history must be connected to grain morphology and phase contrast.

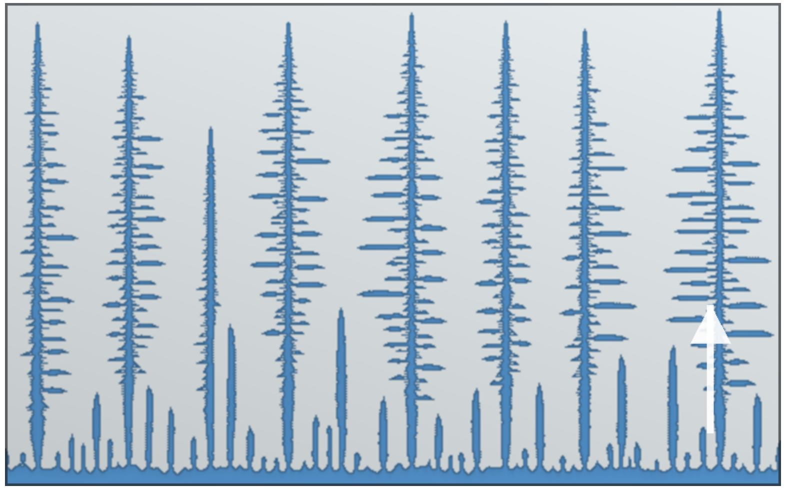
Cellular Automata Solidification Modes

Dendritic



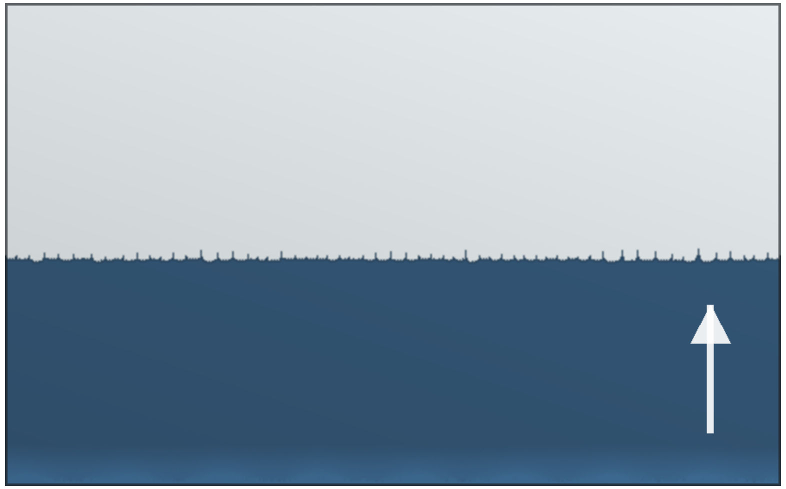
Branching growth when interface perturbations are amplified by undercooling and solute rejection.

Columnar



Competing primary trunks aligned with heat flow select λ_1; sidebranching sets λ_2.

Planar



A stable solid–liquid front advances when the thermal gradient suppresses morphological instability.

Use in this workflow

Eagar–Tsai supplies fast local G and R fields; Bayesian correction should make those fields reliable enough to classify the microstructure.

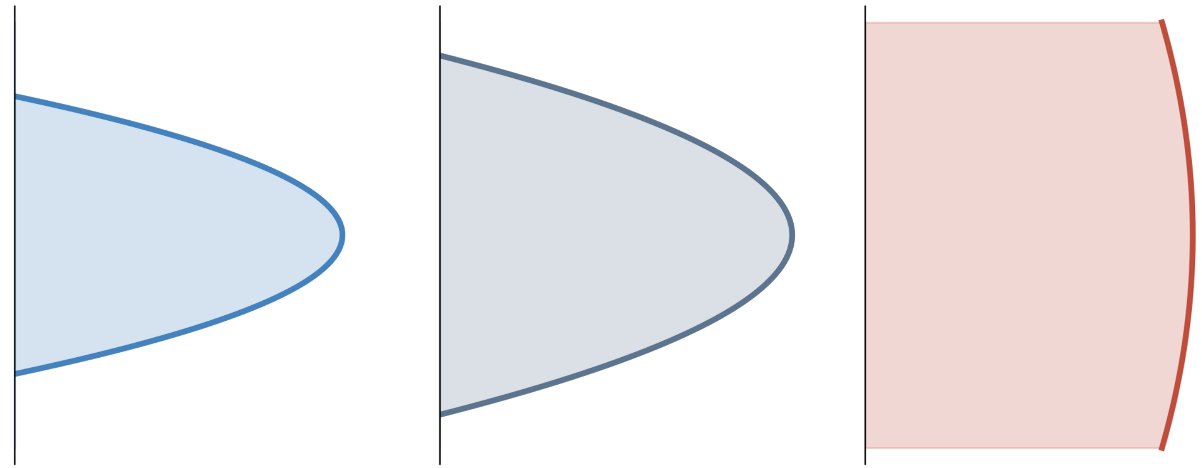
KGT Solidification Theory

Kurz–Giovanola–Trivedi framework

- KGT links dendrite-tip operating state to local thermal gradient, G, and interface growth rate, R.
- The dendrite tip is approximated locally as a paraboloid with radius of curvature ρ.
- Tip selection then relates ρ to R and alloy transport and capillarity properties.

Why it matters here

ET supplies local G and R along the solidification front. KGT uses those fields, together with material properties, to connect melt-pool conditions to dendrite-tip size and morphology selection.



Tip relations

$$z = \frac{x^2 + y^2}{2\rho}$$

$$Pe = \frac{\rho R}{2D_l}$$

$$\sigma^* = \frac{2D_l d_0}{\rho^2 R}$$

$$\rho = \left(\frac{2D_l d_0}{\sigma^* R}\right)^{1/2}$$

$$d_0 \approx \frac{\Gamma}{|m|\, C_0(1-k)/k}$$

- D_l: liquid diffusivity, Γ: Gibbs–Thomson coefficient, m: liquidus slope.
- C_0: nominal composition, k: partition coefficient, σ^*: tip-selection parameter.
- Higher R gives a smaller selected tip radius ρ for fixed material properties.

In this workflow, G determines the imposed thermal field and morphology stability, while R enters directly into the selected paraboloid tip radius.

Eagar–Tsai Model Foundation

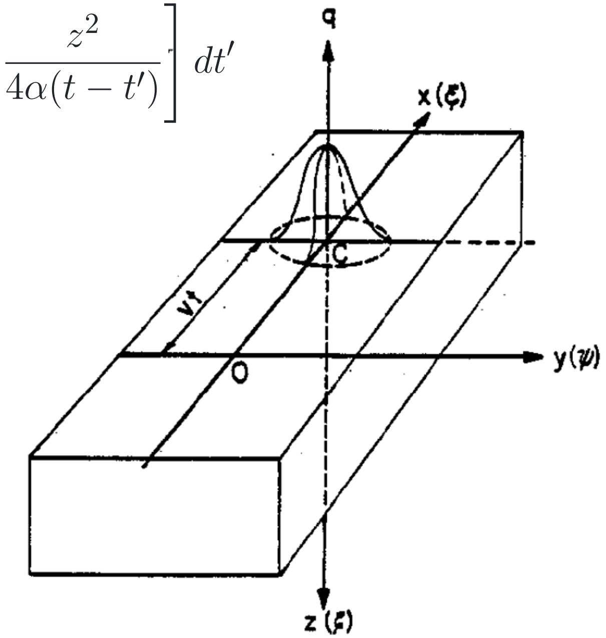
Original Eagar–Tsai Paper

Transient Gaussian moving heat source

$$T - T_0 = \frac{1}{2}\int_0^t dT_i$$

$$= \frac{q}{\pi \rho C_p \sqrt{4\pi\alpha}} \int_0^t \frac{(t-t')^{-1/2}}{2\alpha(t-t') + \sigma^2}$$

$$\times \exp\left[-\frac{(x - vt')^2 + y^2}{4\alpha(t-t') + 2\sigma^2} - \frac{z^2}{4\alpha(t-t')} \right] dt'$$



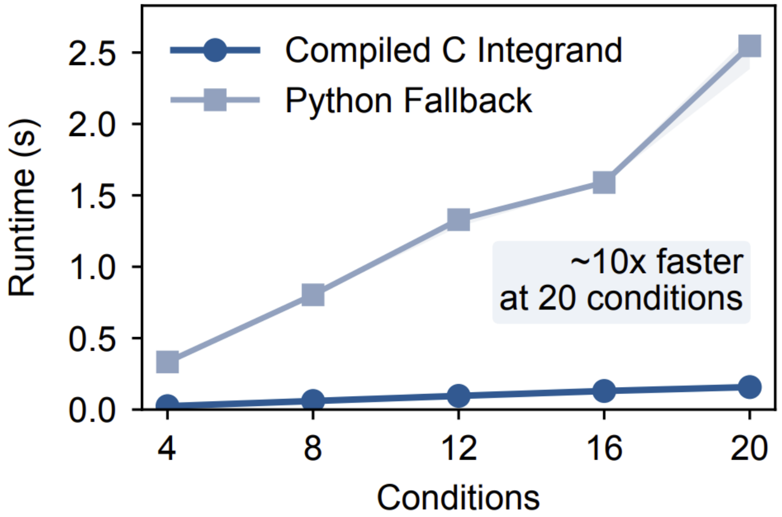
Inputs

Process: P, v, d_b, A
Material: T_L, k_L, ρ_0, C_p

Eagar, T. W., & Tsai, N. S. (1983). Temperature fields produced by traveling distributed heat sources. *Welding Journal*, *62*(12), 346–355.

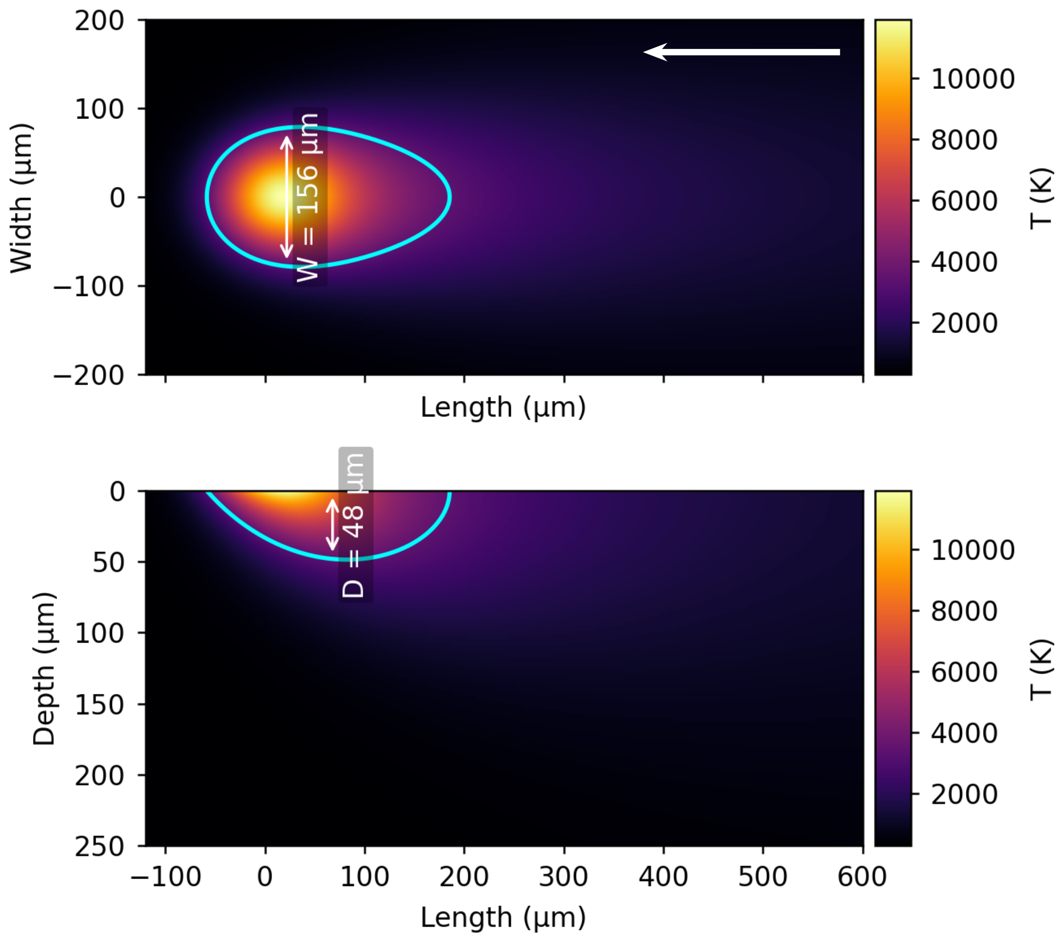
Refactored Eagar–Tsai Model

Python library implementation with compiled-C speedup

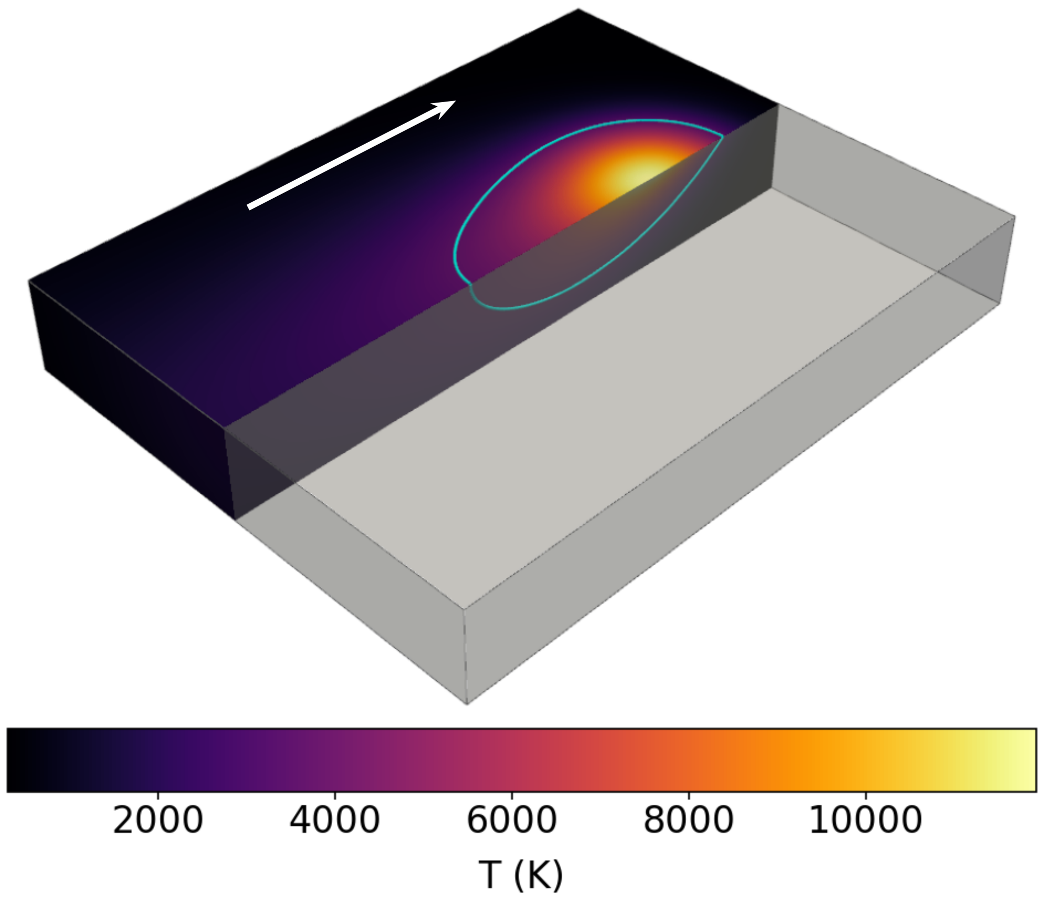


- Separates melt-pool prediction, printability maps, 2D heatmap generation, and 3D temperature-field export into reusable steps.
- Compiled C integrand removes Python overhead inside QUADPACK for faster parameter sweeps.

ET Thermal Cross Sections

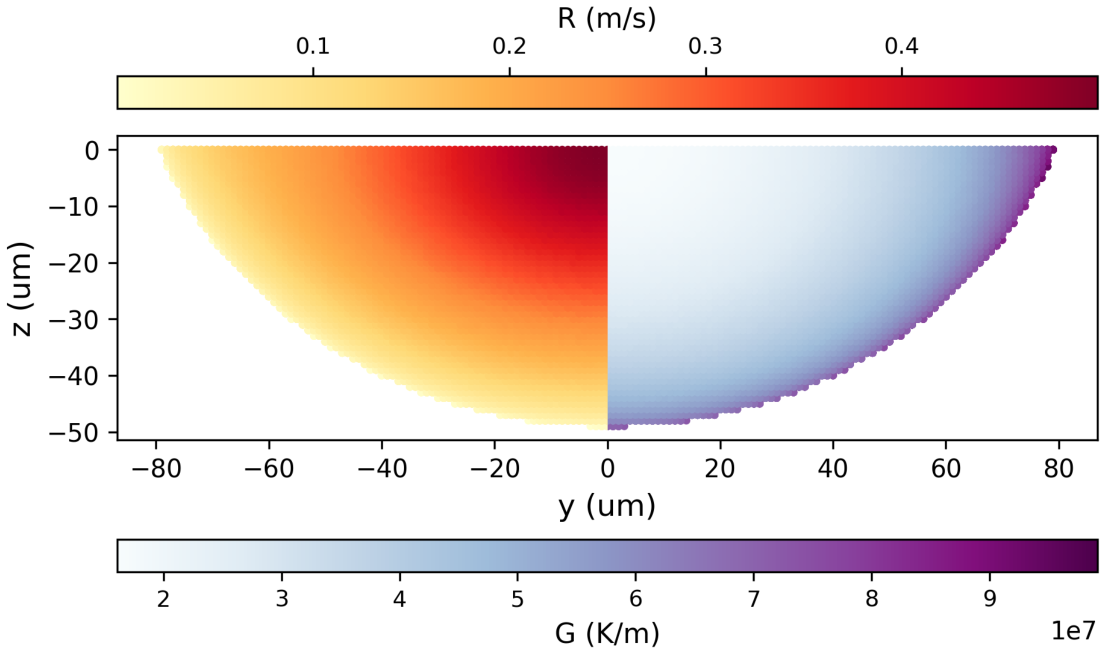


Temperature heatmap for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s. Top: $z = 0$ surface. Bottom: $y = 0$ cross section.

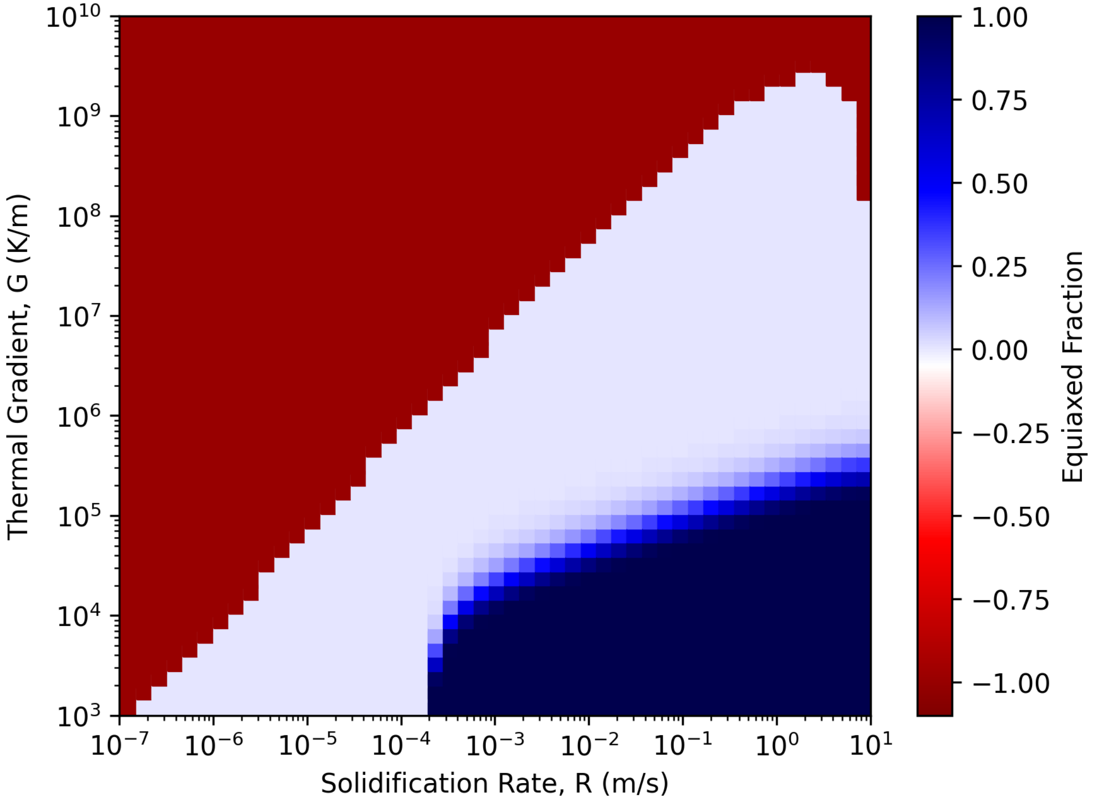


3D view of the ET melt pool for the same alloy and process condition.

ET Thermal Gradient and Solidification Rate

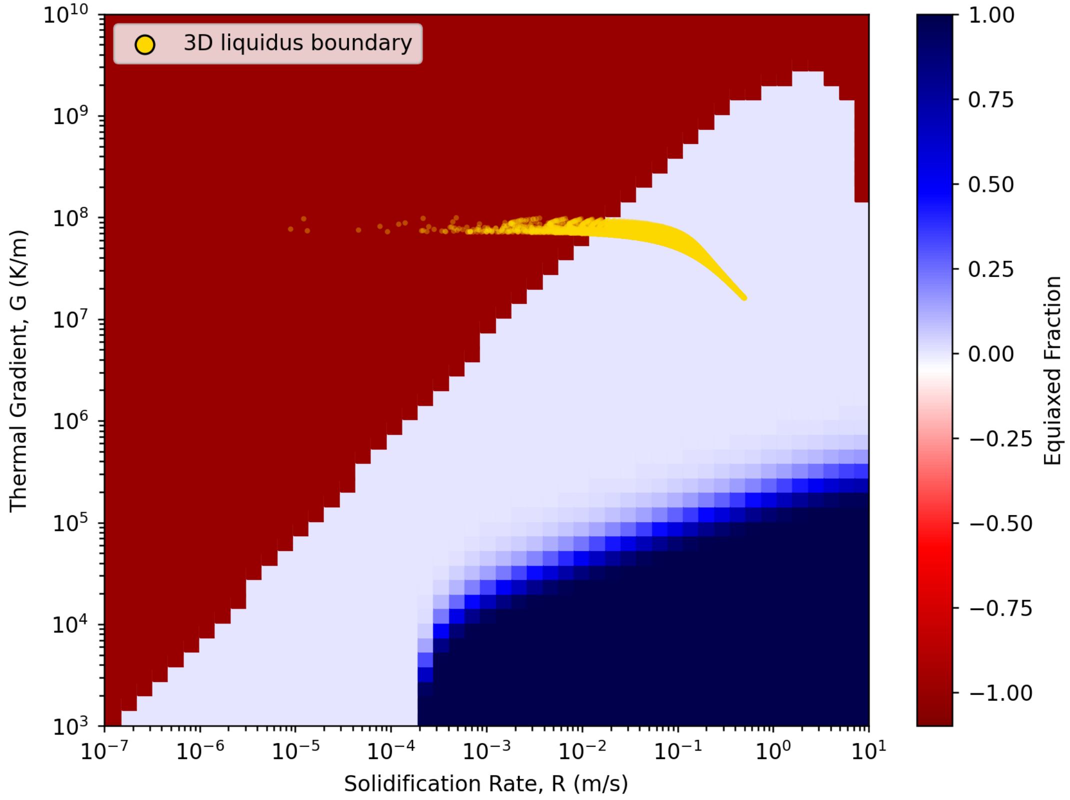


Projection of G and R along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



GR Map calculated with Thermo-Calc CET Property Model for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

ET Microstructure Classification



GR Map calculated with Thermo-Calc CET Property Model with overlaid GR liquidus points for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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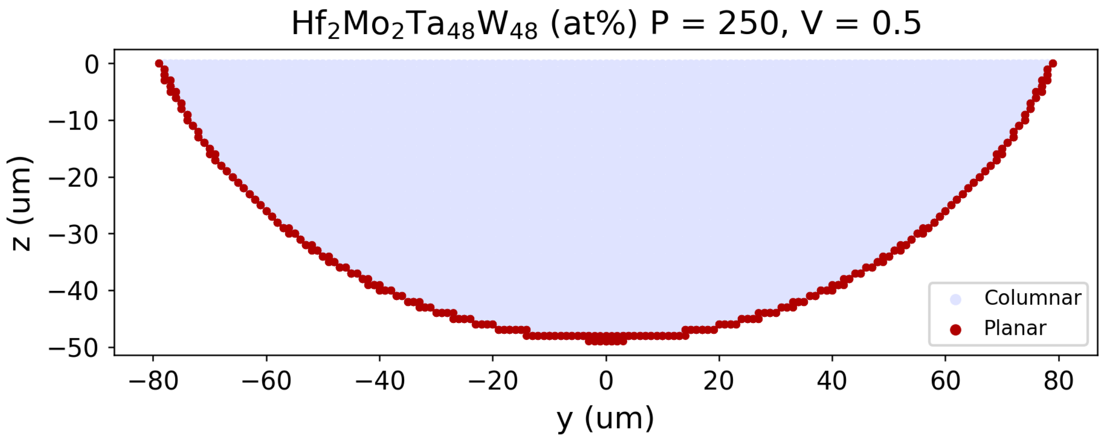
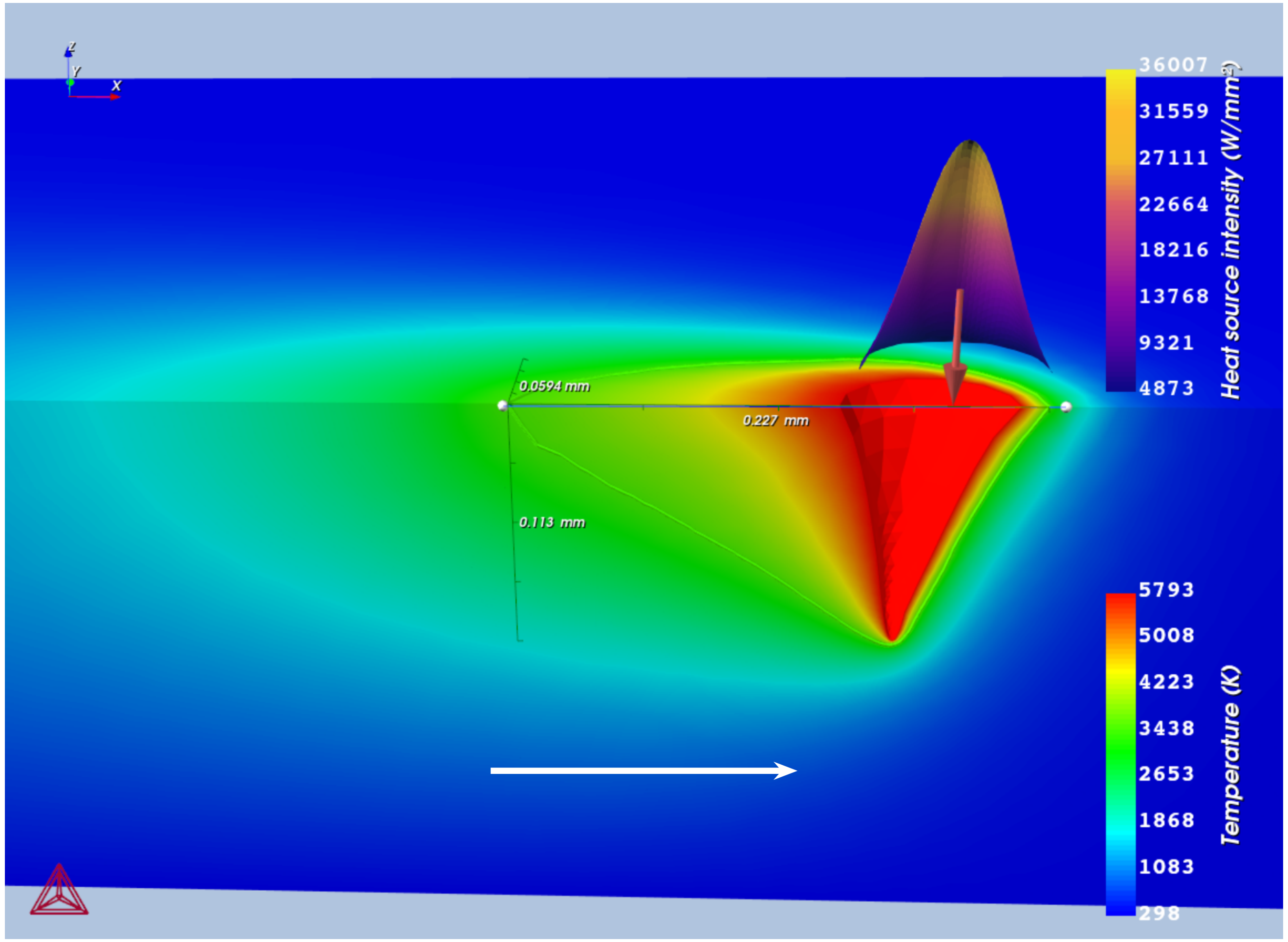


Projected microstructure classification along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

Thermo-Calc AM Module



Steady-state TCAM simulation for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

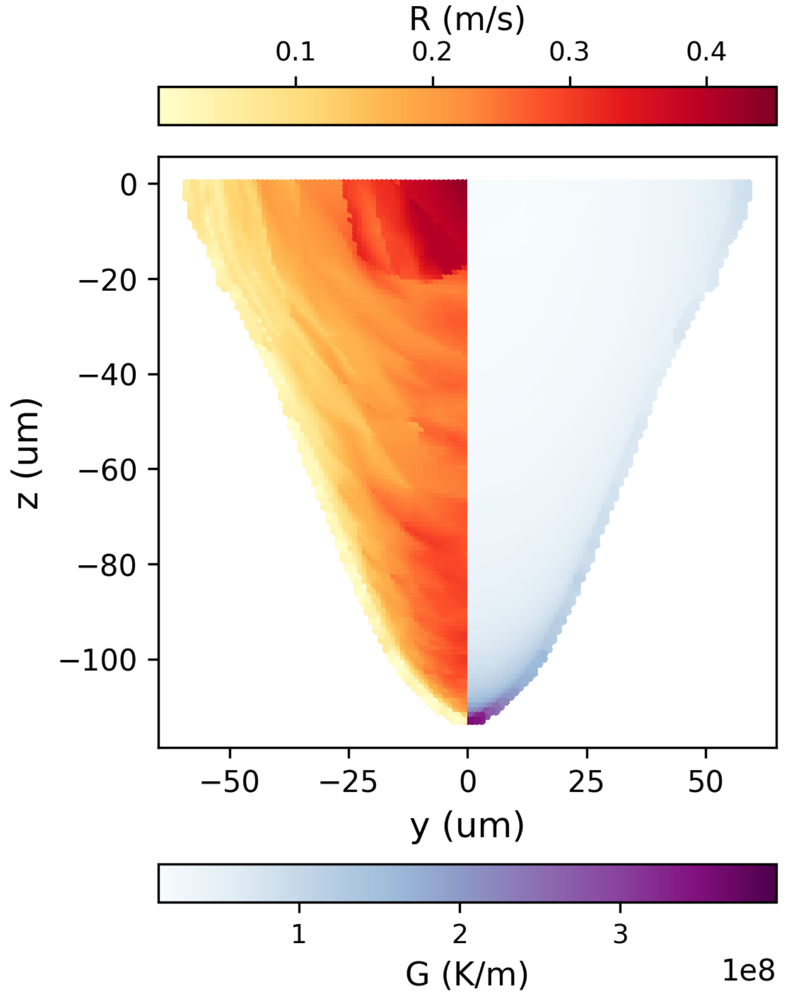
Higher-Fidelity Thermal Reference

- Integrates FEM for thermal conduction with CFD for melt-pool fluid flow.
- Composition-specific, temperature-dependent thermophysical properties.
- Captures Marangoni flow, keyholing, evaporation, radiation, and convection.

Outputs Used in This Work

- 3D temperature field
- Melt-pool geometry
- Thermal gradient (G) and solidification rate (R) from ParaView post-processing

TCAM Thermal Gradient and Solidification Rate



TCAM projection of G and R along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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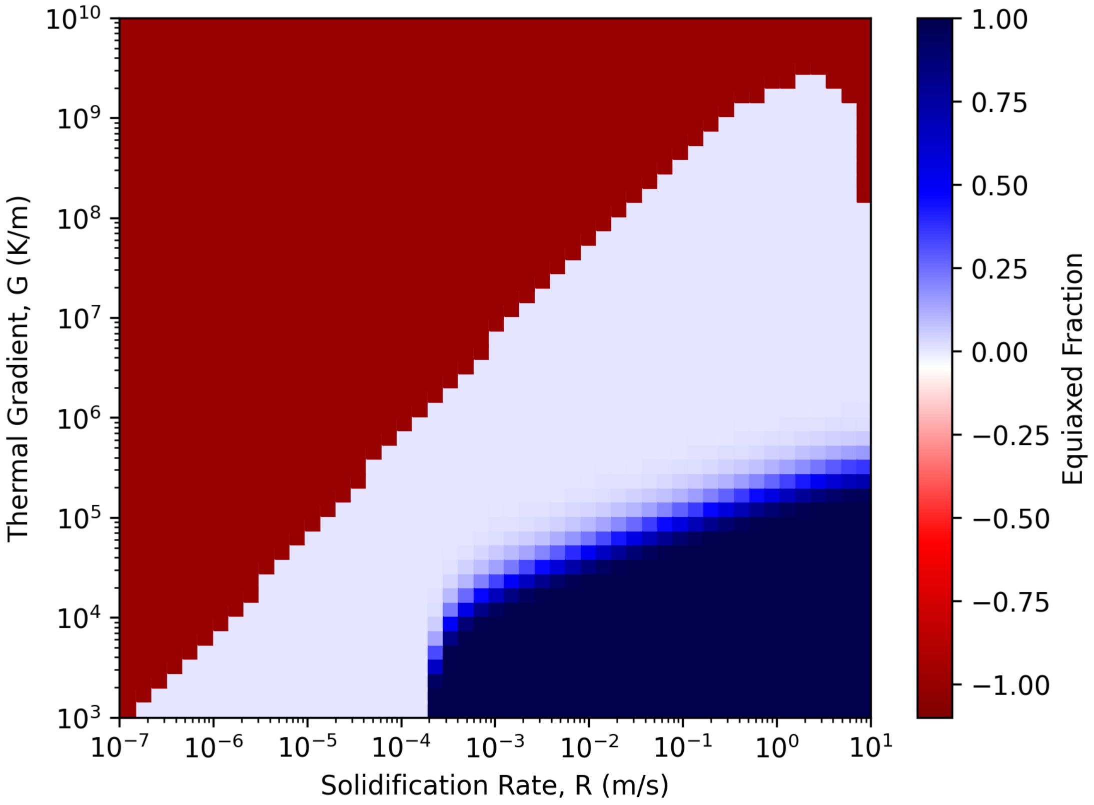
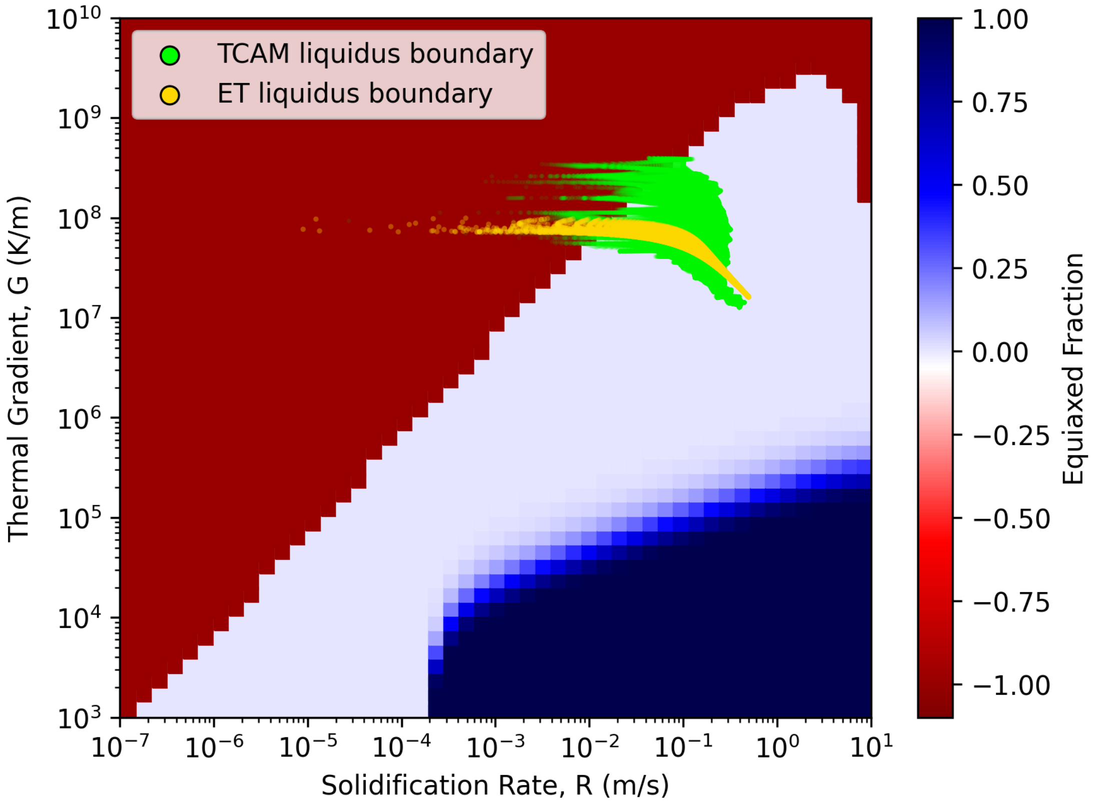


GR Map calculated with the Thermo-Calc CET Property Model for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

TCAM Microstructure Classification



GR map with TCAM and Eagar–Tsai liquidus-boundary conditions overlaid for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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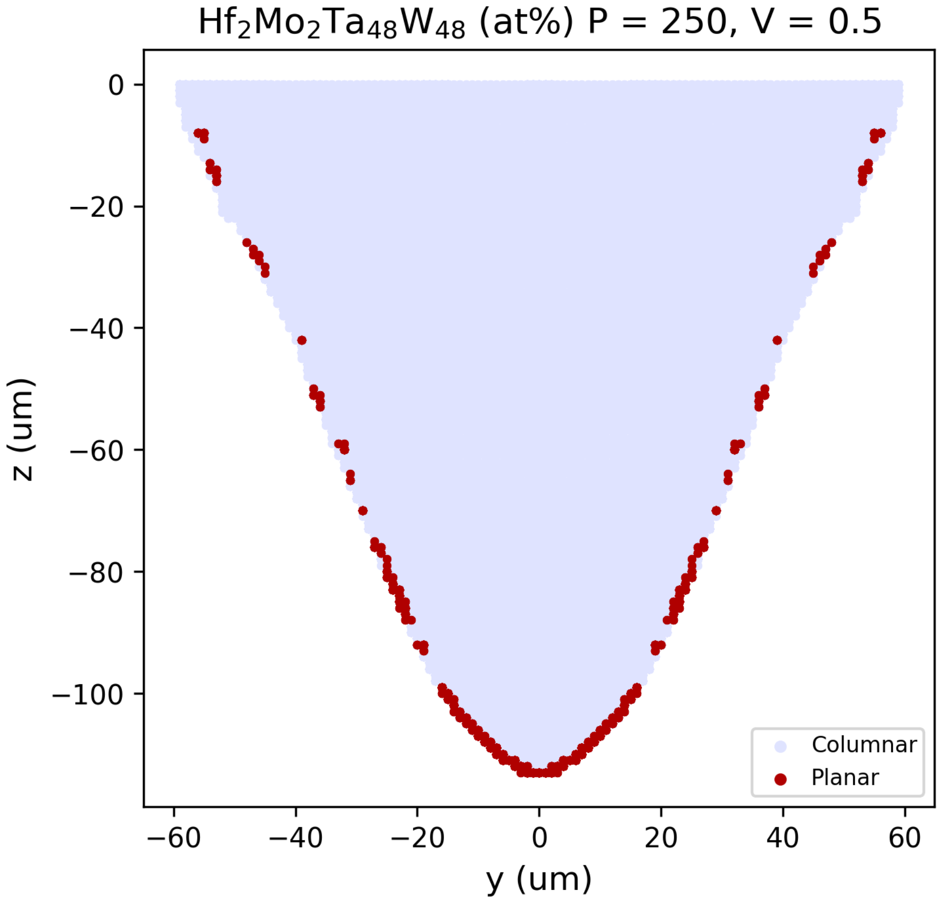
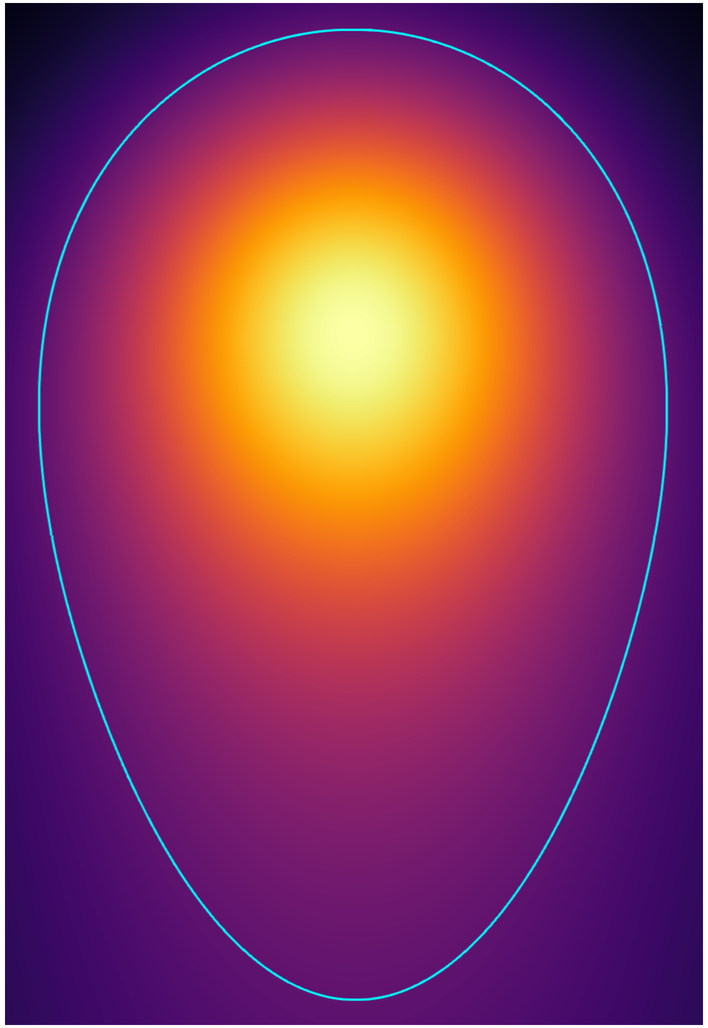
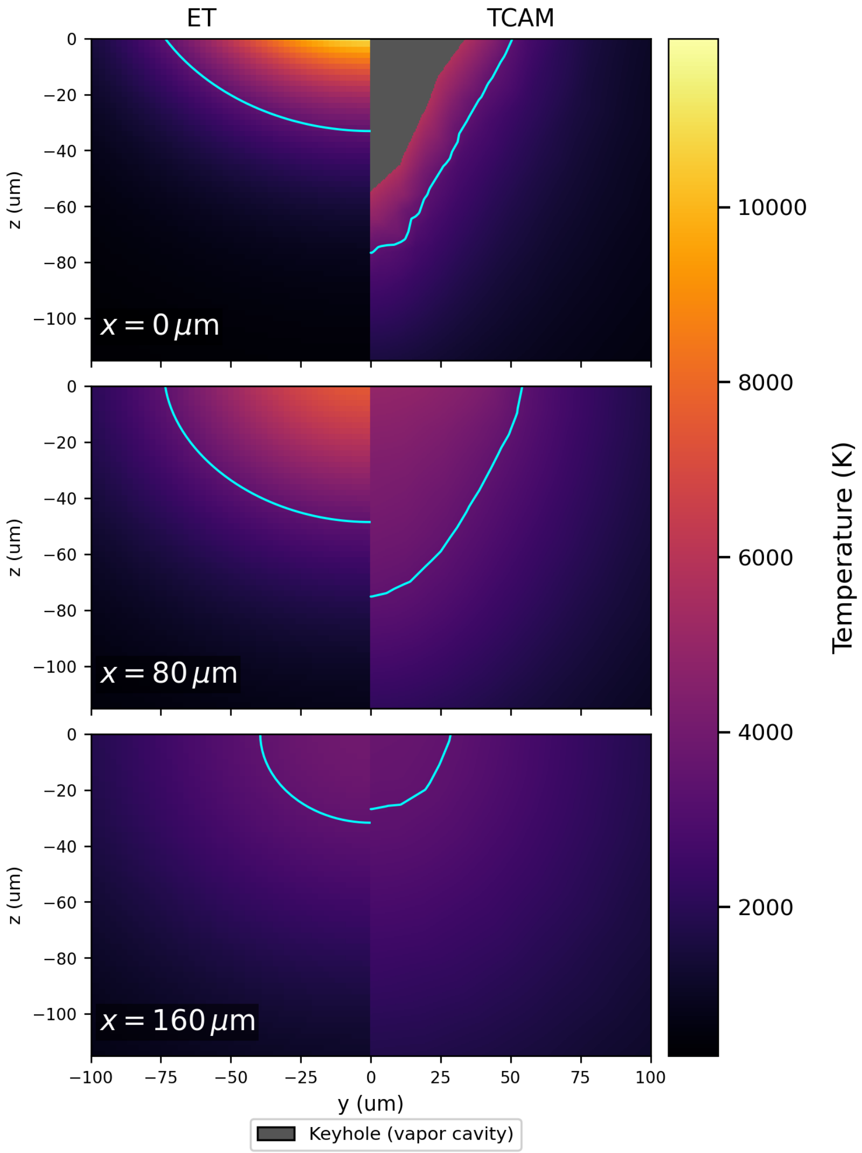
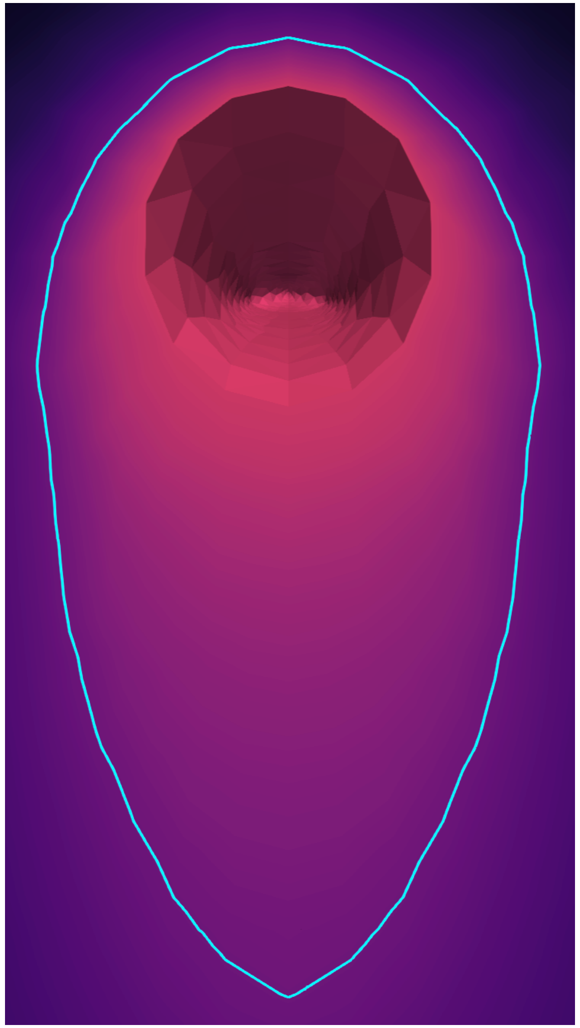


TCAM-derived microstructure classification projected along the solidification front onto a yz slice for the same alloy and process condition.

ET vs. TCAM – Melt Pool Comparison







Bayesian Updating

Physics-informed correction

Eagar–Tsai provides a fast, structured prior across the full melt-pool domain.

Targeted TCAM simulations act as the high-fidelity "ground truth" that teaches the model where and why the prior is wrong.

The correction is learned without discarding the underlying ET physics.

1. ET Physics Prior

Predict temperature rapidly throughout the full melt-pool domain for a spatial location **s** and process condition θ.

$$T_{\text{ET}}(\mathbf{s}, \theta)$$

2. TCAM Training Evidence

Sample high-fidelity temperatures at selected spatial locations and process conditions, then match them to the ET field.

$$T_{\text{TCAM}}(\mathbf{s}, \theta)$$

3. GPR Discrepancy Model

Train the Gaussian process on the structured error rather than replacing the physics model with a purely data-driven predictor.

$$\Delta T = T_{\text{TCAM}} - T_{\text{ET}}$$

Bayesian-Updated ET Model

Posterior Temperature Field

ET supplies the physics-based baseline; the GPR posterior mean supplies the TCAM-informed correction.

$$T_{\text{updated}} = T_{\text{ET}} + \mu_{\Delta T}$$

Spatial Correction

Learns where and by how much ET over- or under-predicts temperature throughout the melt pool.

Process Sensitivity

Across multiple TCAM conditions, learned feature length scales and sensitivity reveal how P, v, d_b, and A affect the thermal field and melt-pool geometry.

Predictive Uncertainty

The posterior variance shows where the correction is trusted and where another TCAM training point would add the most information.

The updated temperature field can then be propagated to melt-pool geometry, G–R conditions, and microstructure classification.

Closing

The same morphology label does not mean the same underlying physics.

Why correction matters

ET and TCAM may both predict “columnar” while disagreeing on:

- Thermal gradient G
- Solidification rate R
- Morphology-transition locations
- Three-dimensional melt-pool geometry

Goal of the Bayesian update

- Preserve ET speed and simplicity
- Recover FEM-informed thermal fidelity
- Propagate the corrected field to G–R conditions and KGT microstructure predictions

Measure of success

- Withheld TCAM conditions
- Accurate temperature fields and melt-pool geometry
- Improved solidification and microstructure predictions